as the distance increases $\left(|\vec{E}| \propto \frac{1}{r^2}\right)$. So the electric field has greater magnitude at the surface A than at B. Therefore, the number of lines crossing the surface A is greater than the number of lines crossing the surface B. Note that at surface B the electric field lines are farther apart compared to the electric field lines at the surface A.

- No two electric field lines intersect each other. If two lines cross at a point, then there will be two different electric field vectors at the same point, as shown in Figure 1.14.

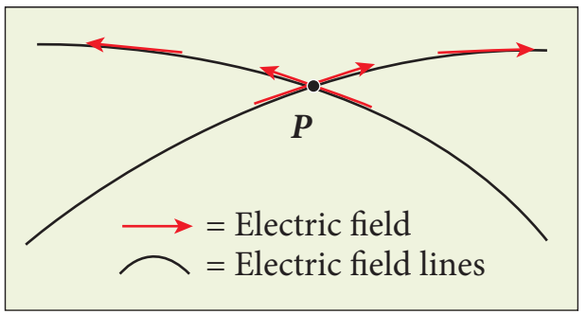


Figure 1.14 Two electric field lines never intersect each other

As a consequence, if some charge is placed in the intersection point, then it has to move in two different directions at the same time, which is physically impossible. Hence, electric field lines do not intersect.

- The number of electric field lines that emanate from the positive charge or end at a negative charge is directly proportional to the magnitude of the charges.

For example in the Figure 1.15, the electric field lines are drawn for charges +q and -2q. Note that the number of field lines emanating from +q is 8 and the number of field lines ending at -2q is 16. Since the magnitude of the second charge is twice that of the first charge, the number of field lines drawn for -2q is twice in number than that for charge +q.

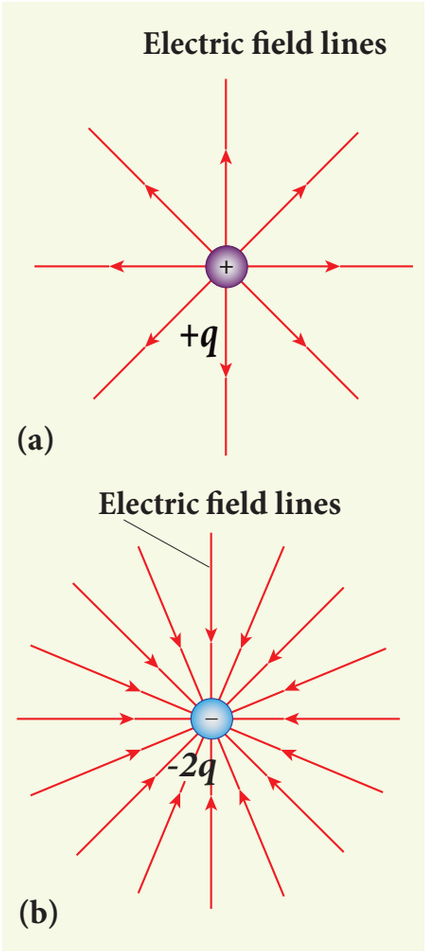
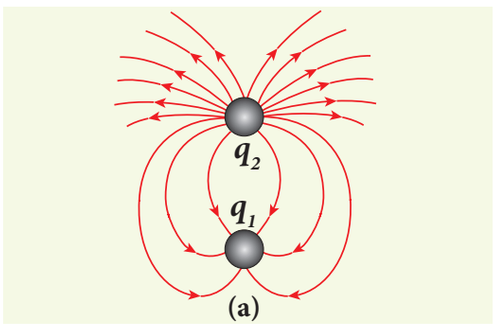


Figure 1.15 Electric field lines and magnitude of the charge

EXAMPLE 1.9

The following pictures depict electric field lines for various charge configurations.



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